PROCESS MINING ASSIGNMENT GUIDE

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GOAL

The goal of this assignment is to:

- Give you the opportunity to apply the acquired knowledge of this course on real data.
- Explore what is possible with modern business analytics techniques and what problems you may face.
- Benefit from the exchange with your fellow students (if you like).
- Obtain a grade for applying your knowledge rather than obtaining a grade for correctly calculating details in a written exam.

GENERAL SETTING



- The general idea is to work in groups of 3-4 students:
  - You may work alone if you prefer (I do not recommend it though).
  - I can help with the group allocation (if desired).
- Submission: **18.10.2026** via Moodle
  Backup: henrik.leopold@nordakademie.de
- Approx. 25 pages for 4 students, 20 for 3 students, 15 for 2 students, 10 for 1 student.
- Keep in mind: You can (and should) use figures / screenshots.

GENERAL SETTING



- The starting point is the provided in two log files:
  - **InternationalDeclarations.xes**
  - **DomesticDeclarations.xes**
- They contain the declarations made for international/ domestic travels at a Dutch university.
- Many details are anonymized but the amounts and the timestamps are roughly correct.
- Your task is to analyze the log files and provide any feedback on the process you can provide to **improve the underlying process.**

TOOLS

Pick **two** out of three:

1. **Process Mining Tools**: I recommend using Celonis or Disco but you are free to use any other process mining tool that you have access to.
2. **Code**: Use a programming language such as Python (e.g. with PM4Py) to analyze the data. Feel free to create and submit a Jupyter Notebook.
3. **GenAI / Agents:** Use GenAI to analyze the data. Note that this is not as trivial as it sounds. You may need to perform some preprocessing of the log (and/or use parts of it only).

Compare the results!

SPECIFIC QUESTIONS / TASKS



- Describe and visualize the general underlying process flow. Make sure you provide a reasonable level of abstraction. You may use BPMN / Petri Nets.
- How long do the different process instances take? Are there any notable patterns?
- Are there any bottlenecks? If yes, where? If yes, can you think of any reasons?
- How many applications get rejected? Can you find any reasons?
- Can you find any unexpected behavior, e.g. missing permits?
- Are there any patterns that would suggest non-conformance?
- Are there notable differences between international and domestic declarations?

SPECIFIC QUESTIONS / TASKS



- What is missing to provide more detailed insights into the processes?
- Reflect on where supervised / unsupervised machine learning techniques could help to obtain further insights.
- Can you think of useful KPIs for the processes?
- Make specific recommendations for improving the processes.

CONTACT

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